MLCB Assignment 1: Epigenetic Age Prediction from DNA Methylation Data

Giorgos Boulogeorgos

March 2026

1 Introduction

DNA methylation—the addition of methyl groups to cytosine bases at CpG dinucleotides—changes systematically with age across the human genome. By measuring this signal with microarray technology, we can train regression models, known as epigenetic clocks, to predict the chronological age of an individual from their methylation profiles. These models have broad applications in biology and medicine, including measuring the rate of biological aging, identifying accelerated aging in various diseases, and understanding the epigenetic basis of human lifespan. The objective of this study is to develop a complete machine learning pipeline to predict chronological age, as well as biological sex, from whole-blood DNA methylation data.

2 Materials and Methods

2.1 Dataset and Preprocessing

This study utilizes the publicly available GSE40279 dataset, which measures whole-blood DNA methylation using the Illumina 450K microarray. The provided dataset consists of a development set (456 samples) and a locked evaluation set (100 samples) containing 1,000 CpG features pre-selected by absolute Spearman correlation with age.

The development data was split into an 80% training set and a 20% validation set, stratified by age using a random seed of 42. To prevent data leakage, a scikit-learn Pipeline was utilized to fit preprocessing parameters exclusively on the training split. Missing CpG values, assumed to be Missing Completely at Random (MCAR), were handled using mean imputation, followed by feature scaling via StandardScaler. Categorical metadata variables (sex and ethnicity) were imputed using the most frequent strategy and encoded via OneHotEncoder.

2.2 Baseline Models

Before feature selection, a plain Ordinary Least Squares (OLS) Linear Regression model without regularization was trained to establish a performance floor. Subsequently, three algorithms were trained at their default hyperparameters on the 1,000 CpG features: ElasticNet (L1+L2 regularized linear regression), Support Vector Regression (SVR) with an RBF kernel, and Bayesian Ridge regression. Validation was performed using bootstrap resampling (1000 resamples, seed = 42).

	Model	Stage	RMSE (mean)	95% CI	MAE	R2	Pearson r
0	OLS	Baseline (Metadata Only)	12.817354	[10.95, 14.90]	10.306695	0.217600	0.486483
1	OLS	Baseline (CpG Only)	7.142080	[4.96, 9.46]	4.642453	0.747522	0.898299
2	OLS	Baseline (Combined)	7.113072	[4.96, 9.44]	4.626398	0.749501	0.898790
3	ElasticNet	Baseline (Defaults)	5.832315	[4.63, 7.29]	4.269786	0.833767	0.916086
4	SVR	Baseline (Defaults)	9.498871	[7.89, 11.29]	7.332972	0.571053	0.804409
5	Bayesian Ridge	Baseline (Defaults)	6.309351	[4.58, 8.44]	4.117792	0.802882	0.912598

Figure 1: *Baseline Models*

2.3 Feature Selection

To improve generalization by removing noisy or redundant probes, two feature selection methods were compared:

- **Stability Selection:** Estimated feature robustness by drawing 50 subsamples (80% of training data). Features ranked in the top 200 by absolute Spearman correlation in over 50% of the resamples were deemed stable.
- **mRMR (Minimum Redundancy Maximum Relevance):** Selected a feature subset of size K by jointly maximizing correlation with age and minimizing inter-feature correlation (redundancy).

2.4 Hyperparameter Tuning & Evaluation

The optimal feature set was carried forward to tune ElasticNet, SVR, and Bayesian Ridge models using RandomizedSearchCV (5-fold cross-validation, 40 iterations). An alternative Bayesian optimization approach using Optuna (Tree-structured Parzen Estimator) was also implemented for comparison. Final tuned models were refitted on the full development set and evaluated on the locked evaluation set using bootstrap resampling.

2.5 Sex Prediction Classification (Bonus B)

To contrast continuous age regression with a concentrated binary signal, sex prediction (Female=0, Male=1) was modeled. Features were re-selected using

mRMR targeting the sex label. Logistic Regression (L2 regularized) and Gaussian Naïve Bayes classifiers were evaluated via 1000 bootstrap resamples on the evaluation set.

3 Results

3.1 Exploratory Analysis

The full development set comprised 456 samples with an average age of 64.45 ± 14.63 years and a relatively balanced sex distribution (49.1% female). The 80/20 stratified split yielded 364 training samples and 92 validation samples, preserving the underlying age and sex distributions.

	Split	n	Age (Mean ± STD)	Sex Balance (% Female)
0	Full Development	456	64.45 ± 14.63	49.1%
1	Training (80%)	364	64.26 ± 14.64	48.4%
2	Validation (20%)	92	65.21 ± 14.67	52.2%

Figure 2: *Dataset statistics table*

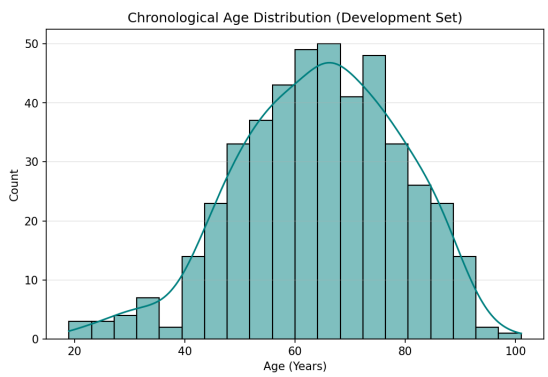

Figure 3: *Age Distribution Figure*

3.2 Feature Selection Outcomes

Stability selection identified 134 robust features across the permutations. For mRMR, candidate values of K were evaluated using an ElasticNet proxy model. Setting $K = 200$ yielded the lowest validation RMSE of 5.7817 compared to

other candidates and outperformed the Stability Selection subset. Consequently, the mRMR features were carried forward.

	Method	Number of Features Selected	Chosen K Justification	Validation RMSE	Validation R^2
0	Stability Selection	185	Fixed threshold (>50%)	6.097308	0.825308
1	mRMR	200	Lowest Validation RMSE among candidates	5.781654	0.842928

Figure 4: *Feature selection comparison table*

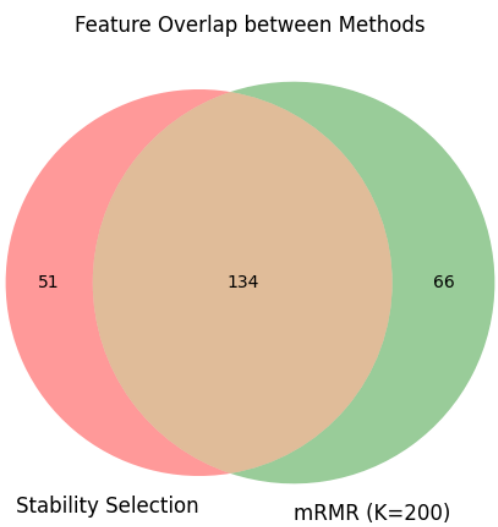

Figure 5: *Venn Diagram*

3.3 Final Model Evaluation (Age)

Post-tuning on the selected features, models were evaluated on the unseen evaluation set:

- **ElasticNet:** RMSE 4.9844 ± 0.7238 (95% CI: [3.76, 6.55]), MAE 3.4515 ± 0.3496, R^2 0.8715 ± 0.0432, Pearson r 0.9360 ± 0.0213.
- **SVR:** RMSE 5.5135 ± 0.6183 (95% CI: [4.41, 6.82]), MAE 4.0812 ± 0.3672, R^2 0.8448 ± 0.0415, Pearson r 0.9222 ± 0.0205.

- **Bayesian Ridge:** RMSE 4.9766 ± 0.7092 (95% CI: [3.77, 6.51]), MAE 3.4868 ± 0.3451, R^2 0.8721 ± 0.0425, Pearson r 0.9367 ± 0.0207.

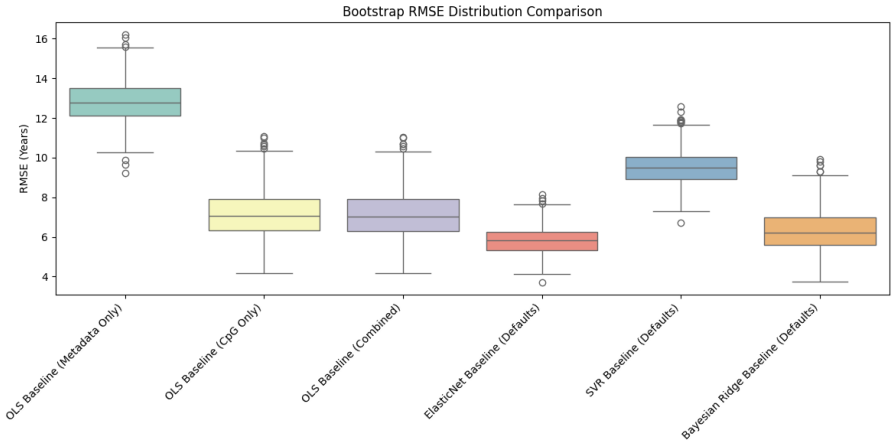

Figure 6: *Bootstrap boxplots(baseline models)*

	Model	Stage	RMSE (mean)	95% CI	MAE	R2	Pearson r
0	ElasticNet	FS+Tuned	4.9844 ± 0.7238	[3.76, 6.55]	3.4515 ± 0.3496	0.8715 ± 0.0432	0.9360 ± 0.0213
1	SVR	FS+Tuned	5.5135 ± 0.6183	[4.41, 6.82]	4.0812 ± 0.3672	0.8448 ± 0.0415	0.9222 ± 0.0205
2	Bayesian Ridge	FS+Tuned	4.9766 ± 0.7092	[3.77, 6.51]	3.4868 ± 0.3451	0.8721 ± 0.0425	0.9367 ± 0.0207

Figure 7: *Final tuned models evaluation table*

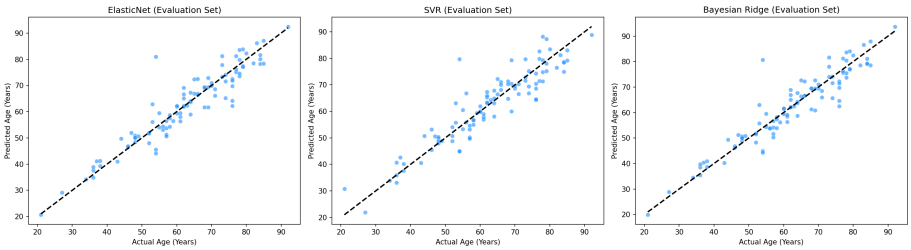

Figure 8: *Bootstrap metrics(tuned models)*

3.4 Optuna Hyperparameter Optimization (Bonus A)

Head-to-head evaluation between RandomizedSearch and Optuna (TPE) was applied to all three models and the results were compared. Both methods suc-

cessfully navigated the hyperparameter space within 40 trials, with Optuna performing marginally better.

	Model	RandomSearch RMSE	Optuna RMSE	Delta	Winner
0	ElasticNet	5.010533	5.008955	1.578411e-03	Optuna
1	SVR	5.523898	5.503328	2.057060e-02	Optuna
2	Bayesian Ridge	5.002008	5.002009	-2.047867e-07	RandomSearch

Figure 9: *Optuna vs RandomsearchCV*

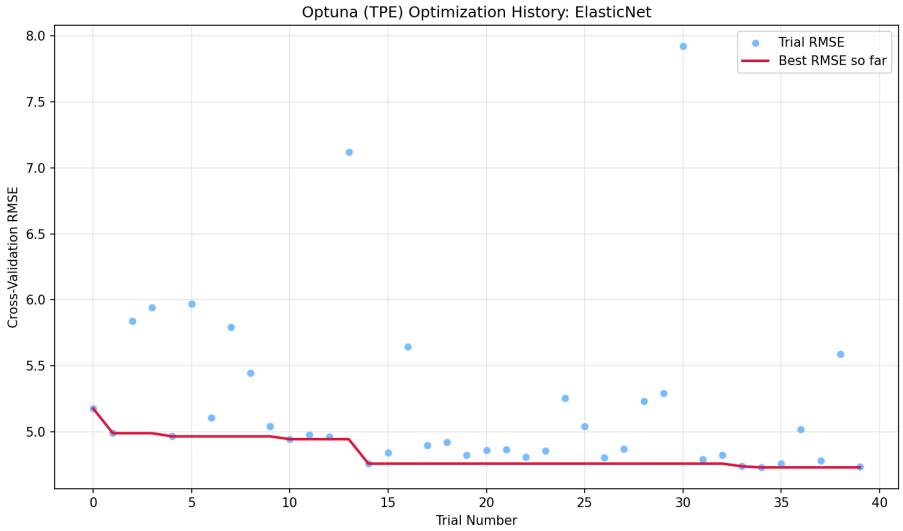

Figure 10: *Optimization History Figure*

3.5 Sex Prediction (Bonus B)

mRMR feature selection targeting the sex label produced a feature set largely distinct from the age-predictive CpGs.

- **Logistic Regression:** Accuracy 0.9405 (95% CI: [0.89, 0.98]), F1 0.9298 [0.87, 0.98], MCC 0.8792 [0.78, 0.96], ROC-AUC 0.9867 [0.96, 1.00].
- **Gaussian NB:** Accuracy 0.6479 (95% CI: [0.55, 0.73]), F1 0.5718 [0.43, 0.69], MCC 0.2744 [0.08, 0.45], ROC-AUC 0.7032 [0.60, 0.80].

	Model	Accuracy (95% CI)	F1 (95% CI)	MCC (95% CI)	ROC-AUC (95% CI)	Features
0	Logistic Regression	0.9405 [0.89, 0.98]	0.9298 [0.87, 0.98]	0.8792 [0.78, 0.96]	0.9867 [0.96, 1.00]	200
1	Gaussian Naive Bayes	0.6479 [0.55, 0.73]	0.5718 [0.43, 0.69]	0.2744 [0.08, 0.45]	0.7032 [0.60, 0.80]	200

Figure 11: *Sex Prediction Table*

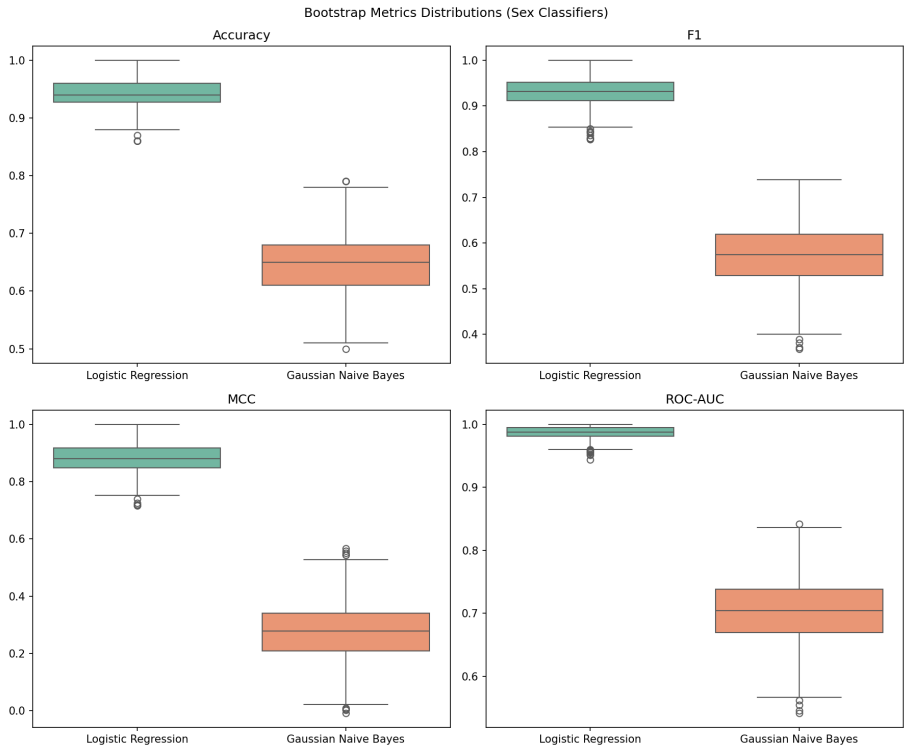

Figure 12: *Bootstrap boxplots*

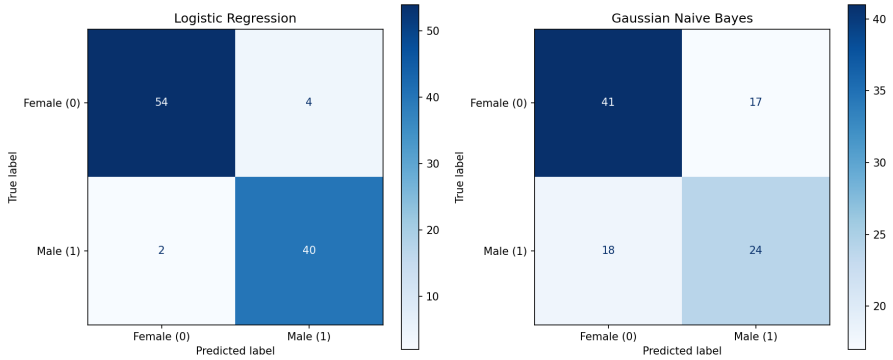

Figure 13: *Confusion Matrices*

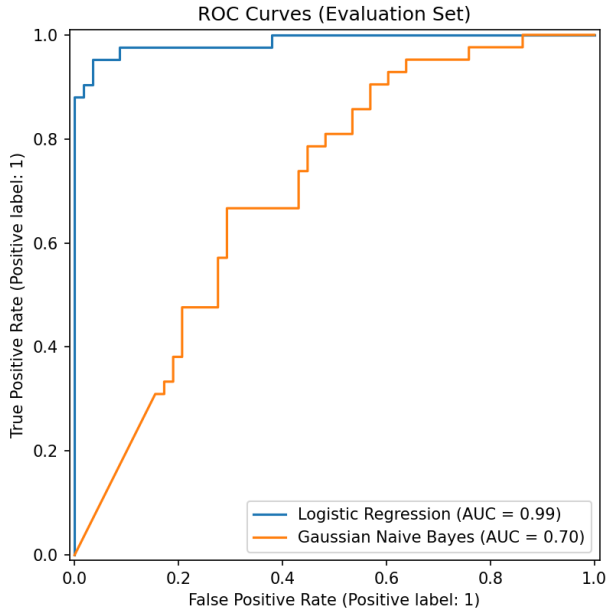

Figure 14: *ROC Curves*

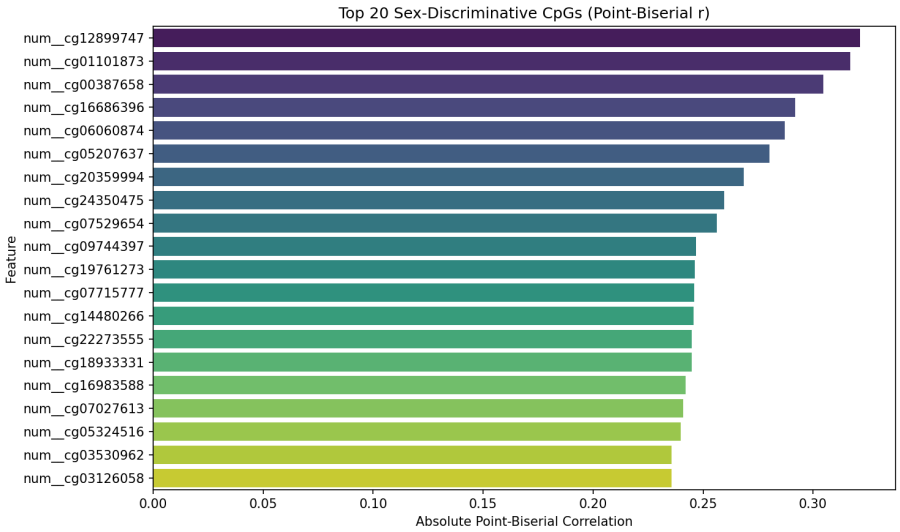

Figure 15: *Top 20 CpGs*

4 Discussion

4.1 Age Model Selection and Performance

Based on the final evaluation, **ElasticNet** is selected as the optimal model for epigenetic age prediction. It achieved the lowest mean bootstrap RMSE and the narrowest 95% Confidence Interval, indicating high reliability. Given the small dataset size, the linear nature and L1/L2 regularization of ElasticNet naturally mitigate the risk of overfitting better than complex non-linear kernels. Furthermore, it provides superior biological interpretability by assigning explicit coefficient weights to individual CpG sites, whereas models like SVR act as a "black box."

4.2 Feature Selection Trade-offs

The mRMR algorithm was favored over Stability Selection. While Stability Selection is highly robust to data perturbations, it ignores inter-feature redundancy. Because DNA methylation data contains highly correlated probes measuring the exact same biological signal, Stability Selection retains redundant noise. mRMR successfully minimized this redundancy while maximizing age-correlation, producing a leaner and more predictive feature set.

4.3 Optuna vs. Random Search

Linear models typically have low-dimensional search spaces (e.g., just `alpha` and `l1_ratio`), meaning random sampling can easily stumble upon near-optimal configurations, which explains why the performance delta was small. However, Optuna's TPE algorithm offers massive theoretical advantages in high-dimensional spaces or computationally expensive models. By building a probabilistic surrogate model, it samples more densely from promising regions, greatly improving sample efficiency over purely random searches.

4.4 Biological Interpretation of Sex Prediction

The feature overlap analysis confirmed that the CpG sites driving the age clock do not overlap with those driving sex differences. Age is a continuous, system-wide drift, whereas sex presents a binary signal heavily mediated by dense methylation near the XIST gene for X-chromosome inactivation in females. Because these X-linked loci are highly correlated with each other, they violate the core assumption of conditional independence required by Gaussian Naïve Bayes. Consequently, Logistic Regression—which utilizes L2 regularization to manage collinearity—produced far superior and better-calibrated results for sex classification.

5 Conclusions

This study successfully constructed a machine learning pipeline to predict chronological age from DNA methylation profiles. By strictly preventing data leakage and employing mRMR feature selection, the high-dimensional 450K array data was distilled into robust epigenetic biomarkers. The tuned pipeline proved to be highly accurate for continuous regression (age) and easily adaptable to binary classification tasks (sex), demonstrating the distinct, highly localized nature of sex-based epigenetic patterns compared to the systemic drift of aging.

AI Use Statement

During the development of this project, a Large Language Model (Google Gemini) was used as a learning aid to assist with debugging scikit-learn pipeline structures and optimizing pandas dataframe manipulations. All biological interpretations, final model selections, and logic flow were critically reviewed, verified, and understood by the author prior to submission.